SRM University *AP*, Andhra Pradesh

Neerukonda, Mangalagiri Mandal,
Guntur District, Mangalagiri, Andhra Pradesh – 522240.

Advanced JAVA Programming

Course Code	CSE 307	Course Category	Core Course (CC)		L	T	P	C
					3	0	1	4
Pre-Requisite Course(s)	OOP with C++	Co-Requisite Course(s)		Progressive Course(s)				
Course Offering Department	CSE	Professional / Licensing Standards						

Course Objectives / Course Learning Rationales (CLRs)

1. To introduce the basic concepts of Object Oriented Programming using JAVA programming.
2. To understand the advanced Java programming concepts.
3. To implement the event handling and database connectivity.
4. To develop Enterprise applications through Java programming.

Course Outcomes / Course Learning Outcomes (CLO s)

	At the end of the course the learner will be able to	Bloom's Level	Expected Proficiency Percentage	Expected Attainment Percentage
Outcome 1	Utilize the Object-Oriented Concepts in solving real word problems through Java.	2	75%	75%
Outcome 2	Develop multithreading applications using Java collections and streams.	3	75%	72%
Outcome 3	Implement GUI based database applications with event-handling mechanisms.	4	75%	70%
Outcome 4	Apply Java programming for enterprise application development.	5	80%	75%

Co urse Art iculatio n Ma trix (CLO) to Pro g ra m Lea rning O utco mes (PLO)

CLOs	Program Learning Outcomes (PLO)														
	Engineering Knowledge	Problem Analysis	Design and Development	Analysis, Design and Research	Modern Tool and CT Usage	Society and Multicultural Skills	Environment and Sustainability	Moral, and Ethical Awareness	Individual and Teamwork Skills	Communication Skills	Project Management and Finance	Self-Directed and Life Long Learning	PSO 1	PSO 2	PSO 3

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Outcome 1	3	3	2	1									3	2	
Outcome 2	3	3	2	1									3	2	
Outcome 3	3	3	3	1								1	3	2	1
Outcome 4	3	3	3	2								1	3	3	1
Average	3	3	2.5	1.25								1	3	2.25	1

Course Unitization Plan Theory

Unit No.	Unit Name	Required Contact Hours	CLOs Addressed	ReferencesUsed
Unit 1	**Overview of Java and Inheritance**	**9**		
	An Overview of Java - Data types, Variables and Arrays, operators,expressions, Control statements	2	1	1
	Classes, Objects, Constructor, Methods, this reference, static keyword, and final keyword Inheritance - Concept, Member access	3	1	1
	Abstract Class, Interface Creating Multilevel hierarchy- super uses, using final with inheritance	2	1	1
	Packages-access specifier	2	1	1
Unit 2	**Polymorphism and Exception handling**	**9**		
	Polymorphism - Compile time Polymorphism, Method overloading, Constructor overloading.	2	1	1
	Runtime polymorphism, Method overriding, Dynamic method dispatch.	2	1	1
	Exception Handling - Fundamentals of exception handling, Uncaught exceptions, using try and catch, multiple catch blocks.	2	1	1
	Exception types - Introduction to Object class, Exception class hierarchy, Built-in exceptions, User defined exceptions, Nested try statements, Throw, Throws, and Finally.	3	1	1
Unit 3	**Multithreading and Even Handling**	**9**		
	Multithreading- Java thread model, Thread life cycle, Creating threads – Thread class, Runnable interface,	2	2	1
	Thread priorities, Synchronizing threads, Inter-thread communication.	2	2	1
	Input/Output - Stream classes, Byte Streams, InputStream, OutputStream, FileIntputStream	2	2	1
	Event Handling - Delegation Event Model, Event classes,KeyEvent class,	2	3	1
	Event Listener Interfaces Using Delegation Event model using AWT, Adapter Classes.	2	3	1
Unit 4	**JDBC and Collections**	**09**		
	JDBC connectivity - Working with SQLite Database,Creation of the Database.	2	4	3
	Connecting to Database, Creation of Tables, Performing various operations(Create,Update,Delete and Retrieve) on tables.	2	4	3
	Collections Overview, Collection Interfaces - List, Set, Map, Collection Classes - ArrayList, Linked List.	2	4	1
	Vector, Set - HashSet, TreeSet Map - HashTable, HashMap, Accessing a collection via an Iterator	3	2	2

Unit 5	**Generic Classes and Advanced GUI Programming with JavaFX**			
	Introduction to Generic Class,Generic Class with Two types of parameter,General Form of a Generic Class,	2	2	1
	UsingWildcardArguments,CreatingGenericMethod,Generic Interface,GenericClassHierarchies.	2	2	1
	JavaFxBasicConcept-AJavaFXApplicationSkeleton-Simple javaFXControl-ExploringJavaFXControls-RadioButton-	2	2	1
	ListView-Combobox-TextField-TreeView-Effectsand Transforms-IntroductiontoJavaFXMenusAnOverviewof MenuBar,MenuandMenuItem-AddImagetoMenuItem-Creating MenuandToolbar	3	2	1

Learning Assessment

Bloom's Level of Cognitive Task		**Internal Continuous Learning Assessments (50%)**					**External Evaluation (50%)**	
		Theory (70%)				**Practical (30%)**	**External Theory (70%)**	**External Lab Exam (30%)**
		CLA-I	**Mid-I**	**CLA-II**	**CLA-III**	**Lab Performance**		
Level-1	**Remember**	**40%**	**40%**	**40%**	**40%**	**40%**	**40%**	**40%**
	Understand							
Level-2	**Apply**	**60%**	**60%**	**60%**	**60%**	**50%**	**60%**	**50%**
	Analyse							
Level-3	**Evaluate**					**10%**		**10%**

Recommended Resources

1. Herbert Schildt (2021). Java: The complete reference, 12th edition, McGraw Hill Education.

2. Java The Complete Reference tenth Edition Herbert Schildt,Oracle Press McGraw Hill Publications.

3. George Reese (2000). Database Programming with JDBC & Java, 2nd edition, OReilly.

4. James C Sheusi (2013). Android Application Development for Java Programmers, Cengage Learning.

5. https://docs.oracle.com/javase/tutorial/jdbc/basics/connecting.html

6. https://www.sqlitetutorial.net/sqlite-java/sqlite-jdbc-driver/

Other Resources

1. E. Balagurusamy (2023). Programming with Java, 7th edition, McGraw Hill.

2. Kathy Sierra, Bert Bates, and Trisha Gee (2022). Head First Java: A Brain-Friendly Guide, OReilly.

3. Horstmann (2023). Core Java Volume I - Fundamentals, Pearson Education, Oracle press.

4. Bill Phillips, Chris Stewart, Brian Hardy, and Kristin Marsicano (2019). Android Programming: The Big Nerd Ranch Guide, Big

Nerd Ranch Guides, OReilly.

5. Valentino Lee, Heather Schneider, and Robbie Schell (2004). Mobile Applications: Architecture, Design and Development, Prentice Hall.

Course Designers

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